Zeyad Hesham

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Summary

Highly skilled Full-Stack Engineer specializing in scalable REST API design, robust backend architecture using NestJS, TypeScript, Node.js, and PostgreSQL, and modern frontend development with React, HTML, CSS, and Tailwind CSS. Proficient in developing production systems with JWT authentication, role-based access control (RBAC), real-time communication via WebSockets, and geospatial querying. Experienced in optimizing database schema design and orchestrating containerized deployments with Docker, ensuring high performance and security.

Skills

- **Domain Skills**: REST API Development, WebSockets, Socket.io, JWT Authentication, RBAC, OpenAPI, Swagger
- **Technical Skills**: TypeScript, JavaScript, SQL, C++, Node.js, NestJS, Express.js, PostgreSQL, TypeORM, Prisma, MongoDB, Mongoose, PostGIS, Docker, Docker Compose, Caddy, Git, React, Electron, HTML, CSS, Tailwind CSS, Shadcn UI, Material UI

Work Experience

Full-Stack Engineer (Freelance) January 2026 - Present

Blue Coast ERP

- Architected a modular NestJS backend with 15+ domain modules including compounds, buildings, floors, apartments, windows, cut plans, glass batches, accessories, pricing, invoices, reports, weights, PDFs, images, and system configuration.
- Engineered a builder-pattern window calculation engine supporting 4 aluminum profile systems (PS 9600, PS 6600, PS 4800, Sonata 45) with configurable sub-types (1-sash, 2-sash, door) and options (fixed profiles, bead types).
- Developed a Best Decreasing Fit bin-packing algorithm for generating optimized aluminum cut plans that minimize material waste, factoring in blade kerf and 45-degree cutting angle adjustments.
- Built a pricing engine that computes per-square-meter window pricing by aggregating aluminum weight, glass area, and accessories costs with configurable waste percentages, plus fabrication, installation, transportation, and profit margin calculations.
- Designed an invoicing module with automatic total calculation from window specifications via integration with the pricing engine, supporting invoice lifecycle management (draft, issued, cancelled).
- Created a 3-tier reporting system (system-wide, compound-level, building-level) with aggregated analytics for material consumption, profile usage efficiency, accessory usage, unit counts, and financial summaries including revenue, gross profit, net profit, and cost breakdowns.
- Generated Arabic RTL PDF reports using pdfmake with Cairo font support, exporting window measurement tables grouped by profile system.
- Implemented JWT-based authentication with access and refresh token rotation, bcrypt password hashing, and role-based authorization (RBAC) enforced globally via NestJS guards.
- Managed database schema evolution with TypeORM migrations and seeded reference data for accessories and profile weights via dedicated seed scripts.
- Containerized the full production stack (NestJS backend, PostgreSQL database, Caddy reverse proxy) using Docker Compose for both development and production environments.

Education

Faculty of Engineering, Alexandria University

Bachelor of Engineering in Chemical Engineering *Graduated: 2025*

Projects

Kitchenly
Designed and implemented a modular REST API with 28+ endpoints for users, menus, orders, and payments using structured architecture and consistent response contracts. Implemented JWT authentication and role-based authorization (RBAC) supporting customer, chef, and admin roles. Integrated Stripe payment processing with secure webhook handling, including idempotency and event validation for reliable transaction processing. Built real-time order tracking using WebSockets (Socket.io) enabling bidirectional state synchronization between clients and chefs. Developed geospatial search using PostGIS for proximity-based chef discovery with optimized radius queries. Designed normalized PostgreSQL schema with Prisma ORM and improved query performance through indexing and optimized relational queries. Applied production-level security practices including rate limiting, input validation, and XSS protection. Generated API documentation using Postman and exported OpenAPI (Swagger) specification for external integration.

Restaurant API
Built a RESTful API for a food delivery platform using Node.js, Express, and MongoDB with structured routing and middleware architecture. Designed consistent response contracts to standardize client-server communication. Implemented JWT authentication and role-based authorization (RBAC) across client, admin, vendor, and driver roles. Modeled scalable NoSQL schemas using Mongoose and implemented core domain operations for restaurants, menus, categories, and orders. Applied security best practices including password hashing, request validation, and input sanitization.

Termadoro
Developed and published a cross-platform CLI productivity tool on npm for Pomodoro-based time tracking with tagging and session analytics. Designed interactive terminal workflows with real-time command controls (pause and resume) and cross-platform desktop notifications. Implemented structured session reporting with tag-based aggregation, history tracking, and export capabilities.

Restaurant Landing Page
Built a responsive landing page using React, Tailwind CSS, and Shadcn UI with a focus on accessibility and responsive design.

Certifications

Mastering 4 Critical Skills Using C++17, Dr. Mostafa Saad Ibrahim — July 2024

Node.js, Express, MongoDB and More: The Complete Bootcamp, Jonas Schmedtmann — February 2025

Harvard CS50 Introduction to Databases with SQL, Harvard — June 2025

The Complete Intro to React Course, — December 2025